

HDOC Day-38

Write a menu-driven C program to perform the following operations using separate functions:

1. Count spaces, digits, and special characters in a string.
2. Count frequency of a given character in a string.
3. Toggle the case of each character in a string.
4. Exit.

Algorithm

Step 1: Start

1. Start the program.
2. Declare a character array `str[100]`, a character `ch`, and an integer choice.
3. Read the string using `fgets()`.

Step 2: Display Menu

4. Display the following menu:
 - **1.** Count spaces, digits and special characters.
 - **2.** Count frequency of a given character.
 - **3.** Toggle the case of each character.
 - **4.** Exit.
5. Read the user's choice.

Step 3: Choice 1 — Count Characters

6. Call the function `countCharacters(str)`.
7. Traverse the string character by character.
8. If the character is a space, increment spaces.
9. Else if it is a digit, increment digits.
10. Else if it is neither an alphabet nor a space, increment special.
11. Display the counts.

Step 4: Choice 2 — Character Frequency

12. Call the function `countFrequency(str, ch)`.

13. Read the character whose frequency is required.
14. Traverse the string.
15. Compare each character with ch.
16. If they are equal, increment count.
17. Display the frequency.

Step 5: Choice 3 — Toggle Case

18. Call the function toggleCase(str).
19. Traverse the string character by character.
20. If the character is lowercase, convert it to uppercase.
21. If the character is uppercase, convert it to lowercase.
22. Display the modified string.

Step 6: Choice 4 — Exit

23. If the choice is 4, terminate the program.

Step 7: Invalid Choice

24. If the choice is not between 1 and 4, display "**Invalid choice**".
25. Repeat the menu until the user selects **4**.
26. Stop.

C Program:

- ```
#include <stdio.h>
```
- `#include <ctype.h>`
  - 
  - `// Function to count spaces, digits and special characters`
  - `void countCharacters(char str[])`
  - `{`
  - `int i;`
  - `int spaces = 0, digits = 0, special = 0;`
  - 
  - `for (i = 0; str[i] != '\0'; i++)`
  - `{`
  - `if (str[i] == ' ')`
  - `spaces++;`

- else if (isdigit(str[i]))
- digits++;
- else if (!isalpha(str[i]))
- special++;
- }
- 
- printf("\nNumber of spaces = %d", spaces);
- printf("\nNumber of digits = %d", digits);
- printf("\nNumber of special characters = %d\n", special);
- }
- 
- // Function to count frequency of a character
- void countFrequency(char str[], char ch)
- {
- int i, count = 0;
- 
- for (i = 0; str[i] != '\0'; i++)
- {
- if (str[i] == ch)
- count++;
- }
- 
- printf("\nFrequency of '%c' = %d\n", ch, count);
- }
- 
- // Function to toggle case of each character
- void toggleCase(char str[])
- {
- int i;
- 
- for (i = 0; str[i] != '\0'; i++)
- {
- if (islower(str[i]))
- str[i] = toupper(str[i]);
- else if (isupper(str[i]))
- str[i] = tolower(str[i]);
- }
- 
- printf("\nString after toggling case: %s\n", str);

- }
- 
- int main()
- {
- char str[100];
- char ch;
- int choice;
- 
- printf("Enter a string: ");
- fgets(str, sizeof(str), stdin);
- 
- do
- {
- printf("\n----- MENU -----");
- printf("\n1. Count spaces, digits and special characters");
- printf("\n2. Count frequency of a given character");
- printf("\n3. Toggle the case of each character");
- printf("\n4. Exit");
- printf("\nEnter your choice: ");
- scanf("%d", &choice);
- }
- switch (choice)
- {
- case 1:
- countCharacters(str);
- break;
- 
- case 2:
- printf("Enter the character whose frequency is to be counted: ");
- scanf(" %c", &ch);
- countFrequency(str, ch);
- break;
- 
- case 3:
- toggleCase(str);
- break;
- 
- case 4:
- printf("\nExiting program...");

- break;
- 
- default:
- printf("\nInvalid choice! Please enter 1 to 4.\n");
- }
- 
- } while (choice != 4);
- 
- return 0;
- }

## Test Case Table:

| Test Case | Input string        | Choice | Additional Input | Expected Output                                |
|-----------|---------------------|--------|------------------|------------------------------------------------|
| 1         | Hello World!<br>123 | 1      | -                | Spaces = 2, Digits = 3, Special characters = 2 |
| 2         | Hello World!<br>123 | 2      | 1                | Frequency of 1 = 1                             |
| 3         | Hello World!<br>123 | 3      | -                | hELLOwORLD!123                                 |
| 4         | Hello World!<br>123 | 4      | -                | Program exits                                  |
| 5         | Hello World!<br>123 | 5      | -                | Invalid choice!<br>Please enter 1 to 4         |

## Compilation & Execution:

The following test cases has been compiled and executed successfully.

Test Case 1:

```
$ cc ndoc38.c
(kali@kali)-[~]
$./a.out
Enter a string: Hello World! 123

—— MENU ——
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 1

Number of spaces = 2
Number of digits = 3
Number of special characters = 2
```

Test Case 2:

```
—— MENU ——
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 2
Enter the character whose frequency is to be counted: 1

Frequency of '1' = 1
```

Test Case 3:

```
—— MENU ——
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 3

String after toggling case: hELLO wORLD! 123
```

Test Case 4:

```
—— MENU ——
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 4

Exiting program...
```

Test Case 5:

```
(kali@kali)-[~]
$./a.out
Enter a string: Hello World! 123

—— MENU ——
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 5

Invalid choice! Please enter 1 to 4.
```